

What these jobs ask for, and what you are missing

Every skill below was counted across the 96 best-fitting engineering roles currently open in the Munich area, drawn from 31 employers. Ranked by how much closing the gap would change your prospects.

Scope: roles scoring 38 or above against your CV and transcripts — tiers 1–3 of the ranked shortlist. Weaker matches are excluded so the priorities reflect jobs you could realistically win.

The short version

- **Functional safety** — asked for by 17 roles across 7 employers
- **CI/CD** — asked for by 13 roles across 7 employers
- **Perception / CV** — asked for by 10 roles across 5 employers

Your control, simulation and PLC foundations are already competitive. What separates you from these postings is not more control theory — it is modern software practice and the safety vocabulary the aerospace employers write into every advert.

1. Gaps worth closing — wanted across many employers

These are ranked by importance: how many roles ask, how many distinct employers ask, and how well those roles otherwise fit you.

	SKILL	JOB	EMPLOYERS	AVG FIT	IMPORTANCE
1	Functional safety Safety, standards & process Do this: Read ISO 26262 part 6 and DO-178C DAL definitions, then re-frame your Roketsan HIL work in safety terms (requirements → test → evidence). A short course certificate is cheap and shows intent.	17 of 96	7 of 31	64	29
2	CI/CD Software engineering practice Do this: Highest effort-to-reward item on this list. Put your thesis Simulink and Python code in GitLab with a pipeline that runs tests and builds on every push. One weekend; visible on your CV forever.	13 of 96	7 of 31	67	27
3	Perception / CV Robotics & autonomy Do this: Pairs naturally with your ROS grade. Working through OpenCV plus a small point-cloud project would open the Munich robotics cluster (NavVis, RobCo, Magazino, Agile Robots).	10 of 96	5 of 31	57	22

	SKILL	JOB	EMPLOYERS	AVG FIT	IMPORTANCE
4	Aerospace standards Safety, standards & process	8 of 96	4 of 31	55	19
	Do this: ECSS and DO-178C vocabulary matters at Isar, RFA and The Exploration Company. Reading the standard summaries costs a day and lets you speak the language in interviews.				

2. Gaps concentrated at a single employer

These look significant by raw count but come from one company's postings. Learn them only if you are targeting that employer specifically — otherwise they are a poor use of your time.

	SKILL	JOB	EMPLOYERS	AVG FIT	IMPORTANCE
1	Distributed systems ONE EMPLOYER Software engineering practice	30 of 96	4 of 31	58	30
	Do this: Mostly Helsing's house requirement. Worth a conceptual grounding (gRPC, message brokers, DDS) only if you target them specifically.				
2	Reinforcement learning ONE EMPLOYER Robotics & autonomy	25 of 96	1 of 31	56	24
	Do this: Entirely Helsing. Ignore unless you are set on their AI roles.				
3	Security clearance ONE EMPLOYER Eligibility & language <i>Requires German residency history; generally closed to non-EU nationals early on.</i>	7 of 96	2 of 31	80	22
	Do this: Not a skill and not something you can prepare for. See the eligibility note — check directly with the employer rather than self-selecting out.				
4	Rust ONE EMPLOYER Languages & programming	14 of 96	2 of 31	63	21
	Do this: Concentrated at Helsing. Do not learn it before the broader gaps.				
5	Machine learning ONE EMPLOYER Robotics & autonomy	14 of 96	3 of 31	56	21
	Do this: Broad but shallow demand in your target roles. A practical PyTorch grounding is enough; you are not competing for research posts.				
6	Kubernetes/cloud ONE EMPLOYER Software engineering practice	8 of 96	4 of 31	51	18
	Do this: Peripheral to control and embedded work. Lowest priority here.				
7	Go ONE EMPLOYER Languages & programming	10 of 96	2 of 31	57	18

3. Partial — you have the theory, not the evidence

The cheapest wins on this document. In each case you already studied the subject; what is missing is something demonstrable a recruiter can see.

	SKILL	JOB	EMPLOYERS	AVG FIT	IMPORTANCE
1	C++ Languages & programming <i>Advanced Programming 1.7 and Simulink-generated code — but no large hand-written C++ codebase</i>	37 of 96	12 of 31	62	43
2	Agile / Scrum ONE EMPLOYER Software engineering practice	16 of 96	4 of 31	57	23
3	Bus protocols Embedded & real-time <i>RS-485 and serial at Phlair; CAN is the common gap for automotive and space</i> Do this: You have RS-485 and serial; CAN is the common missing piece in both space and automotive. A cheap CAN transceiver and a weekend closes it.	9 of 96	6 of 31	60	23
4	Verification & validation Modelling, simulation & test	13 of 96	4 of 31	59	22
5	Orbital mechanics Control, GNC & estimation	4 of 96	2 of 31	80	20
6	ROS 2 Robotics & autonomy <i>Graded 1.0 in Introduction to ROS — but industry has moved to ROS 2</i> Do this: You have ROS 1 at grade 1.0; industry has moved on. Porting one of your lab projects to ROS 2 converts an existing strength into a current one — very high value for the effort.	5 of 96	4 of 31	64	20
7	RTOS Embedded & real-time <i>Echtzeitsysteme 2.0 is coursework; no RTOS shipped</i> Do this: Real-Time Systems 2.0 gives you the theory. Flashing FreeRTOS or Zephyr onto an STM32 turns coursework into something demonstrable.	5 of 96	4 of 31	62	19
8	Software architecture Software engineering practice	6 of 96	3 of 31	63	19
9	Configuration management Safety, standards & process	4 of 96	4 of 31	60	18
10	SQL Languages & programming	7 of 96	4 of 31	50	18

	SKILL	JOB	EMPLOYERS	AVG FIT	IMPORTANCE
11	Kalman / estimation ONE EMPLOYER Control, GNC & estimation <i>Optimal Control and System Identification cover the theory; no shipped estimator</i>	5 of 96	1 of 31	71	18
12	Test automation ONE EMPLOYER Modelling, simulation & test <i>V-Model test harnesses in the semester thesis; no automated regression suite in production</i> Do this: Extend your V-Model thesis work with pytest and a CI runner — this and CI/CD are the same weekend's work.	5 of 96	3 of 31	57	17
13	Linux (embedded) Embedded & real-time	4 of 96	4 of 31	52	17
14	Requirements / MBSE ONE EMPLOYER Safety, standards & process <i>V-Model exposure in the semester thesis; no DOORS or SysML tooling</i> Do this: Named at the primes (Airbus, Leonardo). Learn the DOORS/SysML vocabulary; tool access is usually given on the job.	5 of 96	2 of 31	60	17
15	SCADA / HMI Industrial automation	2 of 96	2 of 31	60	15
16	Bash/scripting Languages & programming	1 of 96	1 of 31	38	9

4. Strengths to lead with

Confirmed in your CV and transcripts, and asked for by these roles. Put these first in your cover letters — they are why you rank where you do.

	SKILL	JOB	EMPLOYERS	AVG FIT	IMPORTANCE
	Embedded development Embedded & real-time	31 of 96	20 of 31	55	48
	C Languages & programming	40 of 96	13 of 31	61	46
	Python Languages & programming	40 of 96	9 of 31	64	42
	GNC Control, GNC & estimation	20 of 96	8 of 31	68	32
	HIL / SIL testing Modelling, simulation & test	20 of 96	5 of 31	65	28

SKILL	JOB	EMPLOYERS	AVG FIT	IMPORTANCE
Git Software engineering practice	11 of 96	6 of 31	74	27
MATLAB Languages & programming	12 of 96	4 of 31	78	26
Simulink / MBD Modelling, simulation & test	10 of 96	4 of 31	82	26
Control design Control, GNC & estimation	8 of 96	7 of 31	62	24
PLC programming Industrial automation	6 of 96	6 of 31	60	22
Instrumentation ONE EMPLOYER Industrial automation	10 of 96	4 of 31	60	21
ROS (general) Robotics & autonomy	7 of 96	5 of 31	61	21
Code generation Modelling, simulation & test	1 of 96	1 of 31	97	21
Docker ONE EMPLOYER Software engineering practice	7 of 96	3 of 31	49	17

5. Eligibility and language — read before drawing conclusions

Your German is not what is holding these roles back

1 of 96 of these roles ask for fluent or C1 German. The Munich deep-tech and aerospace employers you rank well against operate in English, and several say so explicitly. B1 is not the barrier here that it is often assumed to be, and it should not stop you applying to anything on your shortlist.

The honest caveat: that result partly reflects what this tool can see. Employers with public job-board APIs skew toward English-speaking scale-ups; the Mittelstand and public-sector employers who do require German are largely invisible to it. So read this as “German is not gating the roles on your list”, not as “German does not matter in Munich”. Improving it still widens the market beyond what this document covers.

The clearance clause at Isar Aerospace

Isar Aerospace attaches the same sentence to every posting: “Due to security clearance requirements, affiliations with countries listed under § 13 para. 1 no. 17 SÜG may affect the application process.” SÜG is the German Security Clearance Act. The list of states it refers to is maintained by the Federal Ministry of the Interior and is **not public**, so nothing here tells you whether it applies to you.

Two things follow. First, the same adverts state plainly that all qualified applicants are encouraged to apply, and they do not prioritise nationality. Second, this is a question for Isar's recruiters, not for guesswork — ask them directly rather than withdrawing on an assumption. You have already applied to several of their roles, which is the right move.

Airbus differs: its Munich-area adverts require suitability for an *erweiterte Sicherheitsüberprüfung* and very good written and spoken German. Those are stated conditions rather than ambiguous ones.

Mentioned too rarely to prioritise: EU/German citizenship (1), AUTOSAR (3), FPGA / HDL (4), MISRA / static analysis (1), German (fluent) (1) — named by fewer than five roles each, which is too thin an evidence base to act on.

6. How this was measured

- **Document frequency, not word frequency.** A posting naming Python nine times counts once. Raw word counts would have made recruiting boilerplate the top “skill” in this analysis.
- **Employer reach is tracked separately.** Helsing alone writes roughly 40% of the best-fit postings, so its house requirements — Rust, reinforcement learning, distributed systems — would otherwise read as market-wide demand. Section 2 exists to separate the two.
- **Importance** = 45% share of roles + 35% share of employers + 20% average fit of the roles asking. A skill demanded by your strongest matches outranks a commoner one demanded by weak matches.
- **Levels are judgements, not measurements.** They come from your CV, the TUM Leistungsnachweis and the METU transcript. Edit `skills_taxonomy.yaml` and re-run `python3 run.py skills` as they change.
- **Limits.** Only employers with a public job-board API are covered, and a skill absent from an advert is not always absent from the job.